

Multidimensional Goodhart

Response Channels, Scorecards, and Residual Shape

Working draft — six-part spine

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1 Part I — The Licensing Problem

1.1 1. The warning and what it does not identify

Goodhart’s law is usually introduced through the compressed formulation “When a measure becomes a target, it ceases to be a good measure” [20]. The original macroeconomic warning is sharper about policy use: “Any observed statistical regularity will tend to collapse once pressure is placed upon it for control purposes” [7]. Both formulations describe a real empirical phenomenon: once a measure is used for control, the residual error between proxy and target can acquire direction, support, active constraints, exchange rates, tail behavior, and response dynamics. But none of those shapes is identified by the claim that the proxy became a target.

The same observed score improvement can come from several mechanisms. A fixed pool can be re-ranked so better-scoring units enter the selected set. Continuing units can repair measurement errors with no hidden harm. The same units can game a high-score, high-harm action channel. Or the score can improve because the target itself improved. The aggregate score path does not distinguish those stories. A calculation is licensed only after the response model says what is fixed, what changes, and what hidden value or harm is being measured.

This book treats “multidimensional Goodhart” as a response-modeling framework, not as a single universal theorem. The framework asks what proxy pressure changes and which primitives are declared before importing a bound.

1.2 2. What “multidimensional” actually adds

Let S be a state space. A target map $G : S \rightarrow \mathbb{R}^m$ records target-relevant state and a proxy map $P : S \rightarrow \mathbb{R}^k$ records measured features. The intended correspondence is $\varphi : \mathbb{R}^m \rightarrow \mathbb{R}^k$, with residual $\varepsilon(s) = P(s) - \varphi(G(s))$.

Two gaps matter. The **dimension gap** is the part of target variation the proxy map cannot see, represented by directions such as $\ker \varphi$. The **observation gap** is residual measurement artifact inside the measured domain, represented by ε . A scalar score may hide both gaps at once.

Two response channels matter as much as the geometry. A pure selection channel changes participation or selection weights $W_{\theta(u)}$ over a type space U while leaving the response kernel fixed. An intervention channel changes the fixed-type kernel $K_{\theta(ds | u)}$. With baseline type law ν ,

$$\mu_{\theta(A)} = \frac{\int W_{\theta(u)} K_{\theta(A | u)} \nu(du)}{\int W_{\theta(u)} \nu(du)}.$$

Selection and intervention can produce similar marginal score movement. They do not license the same mathematics.

1.3 3. What we tried and what failed

The negative results are not housekeeping; they are the reason the framework has its current shape.

Claim gallery.

- **Unconditional dimensional scaling fails.** Hidden harm does not scale with $\dim(\ker \varphi)$ by itself. If hidden coordinates are independent of the selected proxy, thresholding does not move them.
- **“More measured dimensions means more error” has no sign.** Additive and conjunctive aggregation can give opposite comparative statics from the same primitive action channels.
- **Signed aggregate hidden error is unstable.** Positive and negative hidden movements can cancel. A value functional, norm, tail risk, or domain loss has to be declared.
- **Covariance is not a finite-pressure primitive.** With $P = Z$ and $H = Z^2 - 1$, baseline covariance is zero but threshold or Boltzmann selection moves H .
- **Absolute continuity is not the intervention boundary.** It is useful for the reweighting theorem, but fixed-type action changes can remain absolutely continuous with respect to the baseline law.
- **The selection/intervention split is not marginally identifiable.** It is relative to a declared type/action representation and needs repeated-type, exposure, action-trace, or structural evidence.
- **The selection bound is not coordinate-free by itself.** It is coordinate-explicit until a value norm or scalar value functional is named.
- **Convex affordability is not welfare.** A score-deficit cost bounds private action affordability, not hidden harm.
- **Minimum-complexity attraction is not generic.** Selection follows baseline tails; intervention follows cost, search, caps, fixed charges, and affordances.

These failures block a single “n-dimensional Goodhart law.” What survives is a set of conditional calculations plus a discipline for saying when each calculation applies.

1.4 4. The response-modeling contract

A Goodhart claim must declare the primitives that make it a claim rather than a retrospective label:

Contract.

- **Type representation:** what counts as fixed type u , and what belongs to post-exposure behavior.
- **Baseline behavior:** the baseline type law ν and response kernel K_0 .
- **Policy exposure:** the rule, threshold, ranking, scorecard, or incentive indexed by θ .
- **Response channel:** selection weights W_θ , response kernels K_θ , or an explicit mixture.
- **Action/search geometry:** available actions, costs, caps, fixed charges, search process, and stakes V when an intervention bound is imported.
- **Proxy/target relation:** the maps P , G , φ , residual ε , and the relevant dimension and observation gaps.

- **Aggregation:** additive, conjunctive, threshold, Pareto, lexicographic, or institution-specific rule.
- **Hidden value or harm:** coordinates, scalar value vector, norm, or harm rates h_j .
- **Evidence standard and falsifier:** observations that would distinguish the claimed channel from nearby alternatives and observations that would make the import fail.

The contract is demanding by design. It prevents the framework from inferring welfare, hidden target movement, or a response channel from marginal score movement alone.

2 Part II — Licensed Calculations

2.1 5. Selection channel

This section covers T1 and T2 in `research/core-math.md`.

Before the theorem can be used, the contract must supply a baseline law, a pure selection channel, hidden coordinates or a value metric, and enough integrable moments. Pure selection means $K_\theta = K_0$ and the policy changes only the weights. In the induced marginal law, write $L = d^{\frac{\mu_\theta}{d}} \mu_0$ and $\delta = \|L - 1\|_{L^2(\mu_0)}$.

T1. Coordinate-explicit selection bound. For hidden coordinates H_i with finite second moments and baseline standard deviations s_i ,

$$|\mathbb{E}_{\mu_\theta}[H_i] - \mathbb{E}_{\mu_0}[H_i]| \leq \delta s_i$$

for each coordinate. After a Euclidean coordinate metric is declared,

$$\|B_{H(\theta)}\|_2 \leq \delta \|s\|_2$$

.

T2. Value-weighted/operator selection bound. For a declared scalar value direction v and covariance Σ_H ,

$$|\Delta(v \cdot H)| \leq \delta \sqrt{v^T \Sigma_H v}$$

. For a declared norm,

$$\|B_{H(\theta)}\|_V \leq \delta \sup_{\|v\|_{V,*} \leq 1} \sqrt{v^T \Sigma_H v}$$

.

These are Hilbert-space Cauchy–Schwarz bounds. They license drift envelopes for pure reweighting. They do not identify the hidden coordinates, the value weights, or the welfare object. They do not apply to fixed-type response changes.

Covariance belongs here only as a local velocity. Along a valid exponential tilt, the derivative at zero pressure is a covariance. At finite pressure, the path response, tail shape, and moment-generating domain matter. That is why the zero-covariance example $H = Z^2 - 1$ survives as a warning.

2.2 6. Intervention channel

This section covers T3 and T4 in `research/core-math.md`.

Before an intervention calculation can be used, the contract must supply a fixed-type action model: actions a , private cost $c(a)$, proxy gain $w \cdot a$, score deficit d , and stakes V . The calculation measures the affordability of proxy movement under that model.

T3. Quadratic Stackelberg wedge. In the one-dimensional noiseless threshold toy with quality Q , action $a \geq 0$, private cost $\frac{a^2}{2\kappa}$, threshold t , and stakes V , gaming by a below-threshold unit is privately worthwhile exactly when $t - Q \leq \sqrt{2\kappa V}$.

The wedge $\sqrt{2\kappa V}$ is a signature of this quadratic toy. It is not a universal intervention law and should not be read as an RLHF, benchmark, or organizational theorem unless the action, cost, stakes, and pass condition have been declared.

T4. Convex score-deficit budget. With finite-dimensional action space, closed proper convex cost c , linear proxy gain $w \cdot a$, and regularity for convex duality, define $m(d) = \inf_a \{c(a) : w \cdot a \geq d\}$. Then

$$m(d) = \sup_{\lambda \geq 0} [\lambda d - c^*(\lambda w)]$$

. Gaming under stakes V is feasible exactly when $m(d) \leq V$ in this declared private-cost model.

Cost minimization and hidden-welfare assessment answer different questions. For example, if two proxy channels have equal private cost and equal score weight but hidden harm vector $(M, 0)$, cost minimization splits effort while hidden harm grows with M . The budget licenses private affordability; welfare requires a hidden harm functional.

2.3 7. Multidimensional scorecards — the keeper

This section covers T5 in `research/core-math.md`.

The scorecard calculation needs more than “more metrics.” It needs an additive score, measured channel set M , weights w_j , separable quadratic costs $\frac{a_j^2}{2\kappa_j}$, fixed score deficit d , and linear hidden harm $\sum_{j \in M} h_j a_j$.

T5. Additive exchange-rate iff. The fixed-deficit per-agent hidden harm of the cost-minimizing action is

$$H_{M(d)} = d \frac{\sum_{j \in M} h_j \kappa_j w_j}{\sum_{j \in M} \kappa_j w_j^2}.$$

Fixed-deficit harm is conserved across compared active measured sets exactly when $h_j = c w_j$ on those active channels.

This is the most exportable scorecard result. It says what must be checked before saying whether adding or removing a measured channel helped, hurt, or only re-routed harm: the hidden harm per score unit has to be declared or estimated. Holmstrom–Milgrom multitasking is the closest economics precedent, but the contract here makes the hidden-harm exchange rate explicit.

The result is fixed-deficit and per-agent. Population entry is separate: lowering the private cost of reaching the score can recruit more units into gaming even when

each fixed-deficit gamer has conserved harm. Conjunctive aggregation is separate too; requiring every measured component to clear a bar can make harm grow with the number of components.

2.4 8. Adaptive hardening — narrow but real

This section covers T6 in `research/core-math.md`.

The hardening result is narrow by design. The contract is fixed finite measured set M , fixed deficit d , fixed stakes V , fixed weights, additive proxy gain, separable quadratic costs, deterministic observation, and monotone hardening of capacities $\kappa_{j,t}$.

T6. Deterministic adaptive-hardening capacity boundary. Let

$$S_{t(M)} = \sum_{j \in M} \kappa_{j,t} w_j^2$$

and $T = \frac{d^2}{2V}$. At time t , fixed-deficit gaming is feasible exactly when $S_{t(M)} \geq T$. Hardening reaches no-gaming exactly when $S_{t(M)} < T$. A progress-aware largest-action multiplicative rule terminates in finite time when channels are finite, positive-weight, and floor capacity satisfies $S_{\text{floor}(M)} < T$.

Nothing broader is licensed. The theorem does not cover stochastic observation, arbitrary hardening rules, changing measured sets, changing deficits, changing stakes, nonconvex costs, shared bottlenecks, cycles, welfare comparison, or policy optimality. The immediate anti-transfer is noisy observation: once the update rule can chase noise, the deterministic threshold is no longer a stopping theorem.

2.5 9. Response shape, conditionally

The project did not prove a generic residual-shape law. What survives is a conditional taxonomy:

- Quadratic costs select the minimum-cost direction, often proportional to Cw in a smooth unconstrained model.
- Fixed charges create entry thresholds and active-set comparisons.
- Caps produce spillover only after a channel saturates.
- Low-rank affordances constrain response to an image, but do not choose a basis-invariant “simple” direction by themselves.
- Search-prior claims require a coding language or search process fixed before the failure is observed.

These are not theorem transfers between domains. They are ways to turn a response-shape conjecture into a declared model with falsifiers.

3 Part III — Across Disciplines: Primitive Attribution

3.1 10. Genealogy

Goodhart, Campbell, Strathern, and Manheim–Garra-brant are genealogy, not proof sources for the calculations above. Goodhart’s original macroeconomic warning concerns policy-contaminated regularities [7]. Campbell’s warning concerns social indicators under decision pressure [2]. Strathern’s formulation supplies the familiar compressed warning [20]. Manheim and Garra-brant provide a useful cause taxonomy [13].

This book does not replace that genealogy with a grander taxonomy. It uses the genealogy to ask which primitives are present: selection, response kernel, action cost, proxy/target separation, aggregation, hidden value, and evidence standard.

3.2 11. Formal analogues, primitive by primitive

The closest formal analogues supply some primitives and omit others. El-Mhamdi–Hoang and Majka–El-Mhamdi are scalar tail-conditioned selection anchors [5, 12]. Hardt et al. strategic classification supplies costly feature-change intervention [8]. Perdomo et al. performative prediction supplies deployment-induced response kernels [15]. Skalse et al. supply proxy/target and RL-specific response-kernel tools [11, 18]. Holmstrom–Milgrom supplies the closest multitask incentive analogue for aggregation and action costs [10]. Cawley–Talbot, reusable holdout, leaderboard work, and optimizer’s curse supply selection and evidence standards [1, 3, 4, 16, 17, 19]. Pan and Gao provide empirical response-geometry candidates, not plug-in parameters [6, 14].

3.3 12. What each discipline contributes vs. omits

Source family	Primitives supplied	Primitives omitted	Licensed transfer
Tail-conditioned Goodhart	Selection, scalar proxy/target, tail assumptions.	Vector aggregation, intervention, hidden welfare.	Sharp scalar asymptotics where the tail contract holds; otherwise only a reweighting analogy.
Strategic classification and performative prediction	Response kernels, costly action or endogenous deployment.	General hidden value, scorecard exchange rates, marginal identification.	Intervention framing after action/cost or deployment response is declared.
RL reward hacking	Proxy/target separation, MDP oc-	Non-MDP domains, empirical	RL-specific response calculations

Source family	Primitives supplied	Primitives omitted	Licensed transfer
	cupancy geometry, optimization pressure.	welfare, generic cost parameter.	only inside the MDP/occupancy contract.
Multitask incentives	Aggregation and multidimensional effort/cost.	The book’s value-metric contract, non-LEN settings unless separately shown.	External precedent for the score-card exchange-rate question, not a proof of this book’s theorem.
Adaptive holdout and leaderboard hygiene	Selection/evidence standards for repeated benchmark use.	Fixed-type fine-tuning, contamination, welfare, action cost.	Evidence discipline for selection over fixed candidates and public/private score interpretation.
Reward overoptimization empirics	Candidate search and optimization-pressure variables.	A defended mapping from model size, KL, data access, or prize to κ or V .	Hypothesis generators for response geometry; no automatic theorem transfer.

3.4 13. Reduction, not unification

No single equation unifies these literatures. The framework’s claim is narrower: each formalism fills different fields of the response-modeling contract. The contract says which primitive is being borrowed and which transfer fails. A source about adaptive holdouts does not identify hidden welfare. A source about costly feature change does not identify value weights. A source about multitask contracts does not make “more metrics” good or bad in a new application.

4 Part IV — Cases as Stress Tests

4.1 14. MMLU

MMLU is useful because it is not a single mechanism [9]. A public benchmark score can induce fixed-checkpoint model selection, repeated adaptive testing, prompt search, finetuning on benchmark-like data, contamination, reward-model overoptimization, or genuine capability transfer.

The contract separates these mechanisms. Fixed-checkpoint comparison is a selection problem: candidate checkpoints are types, the score is a proxy, and the hidden target is performance away from the benchmark. Finetuning, prompt search, contamination, or reward optimization is intervention/search only after an action/search geometry is declared. MMLU alone supplies neither κ nor stakes V , and it does not supply hidden welfare. Proposition 3 applies only after a cost model and score deficit are declared. Proposition 4 applies only after benchmark components are modeled as additive channels with harm exchange rates.

The stress test is whether the evidence can distinguish fixed-candidate selection from post-exposure adaptation. Necessary observations include benchmark access, training data lineage, prompt/search budget, repeated-query history, transfer to non-MMLU tasks, and failure modes that move when the benchmark format changes.

4.2 15. Hospital readmissions

For a readmission scorecard, score improvement is compatible with hospitals leaving the comparison pool, coding changes, observation-status changes, delayed admissions, patient avoidance, better follow-up care, or mixtures.

The auditable primitives are concrete: effective score weights w , stakes V , response ease κ for coding, discharge timing, follow-up, and patient selection channels, hidden harm rates h , and signal adequacy γ for the measured components. If these are unavailable, the design consequence is not “assume the score is bad.” It is: do not use the toy diagnostic, collect action traces, monitor hidden patient outcomes where possible, pilot or lower leverage, and mark the missing primitive.

This is an evidence contract, not policy advice. The framework can say that aggregate readmission movement is insufficient to credit patient-welfare improvement. It cannot rank hospital policies without the clinical and value model the contract explicitly requires.

4.3 16. Scientific metrics

Publication counts, citations, grants, venue prestige, and rankings are already covered by responsible-metrics warnings such as DORA and the Leiden Manifesto. The framework does not claim novelty for the warning that crude metrics can distort science.

The stress test is discriminator value. A scientific scorecard application must separate selection over researchers, labs, fields, and institutions from fixed-researcher response. It must distinguish harmful proxy manufacture from harmless metadata/

discoverability repair and from genuine improvements in methods, robustness, replication value, useful synthesis, or long-run uptake.

The contract refuses broad research-value inference from score movement. If an institution cannot declare the target, aggregation, action traces, and hidden value model before interpreting a score rise, the framework returns “no licensed hidden-value claim.”

5 Part V — What the Framework Refuses to Do

5.1 17. Anti-applications

The framework is least useful when its primitives cannot be stably declared. Concrete anti-applications include:

- A rapidly shifting online platform where type space, measured set, policy exposure, and available actions all change between measurements.
- A black-box institutional ranking where effective weights and thresholds are unobservable and participants cannot tell which action channel is being rewarded.
- A high-stakes public decision where stakeholders demand a welfare verdict but no defensible hidden value model exists.

In these cases, the contract returns “no verdict.” That is not a hedge; it is the framework refusing to convert score movement into causal mechanism or welfare language without the primitives that would make the claim testable.

5.2 18. Falsifiers

The contract itself is falsifiable as a modeling discipline. A concrete falsifier would be a domain where the primitives are declared in advance, response channel and action traces are observable enough to distinguish the nearby mechanisms, and the licensed calculation systematically predicts the wrong response shape or direction while a simpler score-only rule predicts it correctly.

Examples include:

- In a pure-selection setting with measured L and hidden coordinates, hidden drift repeatedly exceeds the δs_i envelope under the stated moment and absolute-continuity conditions.
- In a fixed finite deterministic hardening setting, with stable M , d , V , weights, separable quadratic costs, and deterministic observation, gaming remains feasible after $S_{t(M)} < \frac{d^2}{2V}$ or stops while $S_{t(M)} \geq \frac{d^2}{2V}$.
- In an additive scorecard with defended w_j , κ_j , and h_j , fixed-deficit per-agent harm is conserved across active measured sets even when the exchange-rate condition $h_j = cw_j$ fails, or fails to be conserved when the condition holds.

If those failures survived audit, the framework would not merely need prose repair; the licensed theorem import would be wrong or the contract fields would not track the response mechanism they were supposed to track.

6 Part VI — Open Agenda

6.1 19. The residual-shape conjecture

The signature open problem is Q18: when does repeated proxy repair drive hidden failure toward a predictable residual shape, such as a low-complexity attractor? The current answer is negative unless a mechanism is named first.

Resolving Q18 requires at least five declarations: a response mechanism, a complexity or shape functional fixed before inspection, a policy-update rule, a composition rule for repeated repair, and a failure condition. It would count as progress to prove a sparse, low-rank, or low-description-length attractor inside one of those contracts. It would not count to observe a simple-looking failure after the fact and relabel it as the attractor.

6.2 20. Composition, identification, and information

Three structural gaps remain.

First, channels compose. Real cases mix selection, fixed-type response, proxy repair, and real improvement. The current contract names mixtures but does not give a general calculus for them.

Second, primitives are not automatically identifiable. Marginal score movement does not identify W_θ versus K_θ , hidden harm, action costs, or aggregation. The framework needs identification toys and evidence thresholds for when a primitive can be treated as declared rather than guessed.

Third, the L^2 selection bound may not be the portable final form. An information-theoretic restatement could travel further if it preserves the distinction between coordinate-explicit drift, declared value metrics, and finite-pressure path behavior.

6.3 21. Toolkit gap

A practitioner should not have to re-derive the contract every time. The missing toolkit has three parts: a primitive-elicitation protocol, worked exchange-rate audits in real scorecards, and small identification examples that show what observations distinguish selection, intervention, proxy repair, and real improvement.

Until those tools exist, the framework is best read as a claim-license discipline and a theorem inventory. It tells a reader what would have to be true before a Goodhart calculation travels.

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